

Hindsight-Supervised Kolmogorov-Arnold Gates for Interpretable Control of Image Classifier Training

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ABSTRACT

Training a deep convolutional network for image classification routinely consumes compute long after validation performance has stopped improving, yet the decision to stop a run or lower its learning rate is still made by eye or by schedulers whose reasoning cannot be inspected. We present an agentic controller for vision model training in which a Kolmogorov-Arnold Network (KAN) serves as an interpretable decision gate. A monitoring pipeline observes a live CNN training process and reduces its validation-loss and gradient history to five normalised features. A language-model supervisor proposes an action, but the decision that is executed (continue, halve the learning rate, or stop) is always the KAN gate's. After training, every edge of the gate is fixed to a closed-form function, so the formula shown to the operator is exactly the function that produced the decision; it reproduces the network output to within 6.5×10^{-4} on a 0 to 100 score scale. A dashboard streams the run, the agent steps, and each decision traced through the learned edge functions.

Our main methodological contribution is hindsight supervision. Training the gate to imitate a hand-written rule limits it to that rule, and relabelling live runs with the gate's own decisions is circular. Instead, every decision point of a completed run is labelled by the validation-loss improvement that actually remained, so the gate learns from outcomes rather than from a teacher.

We evaluated the system on 36 complete, real training runs of a CNN on Fashion-MNIST images, covering six learning rates, two batch sizes, and three seeds with a 40-epoch budget. Evaluation used cross-validation grouped by seed, so no test run shared an initialisation with any training run. The hindsight-trained gate saved 31.0% of the epoch budget at 0.20% mean regret in best validation loss, close to the 0.18% of an oracle that knows the future. The same gate trained on rule labels saved only 4.4%. Its savings are comparable to a well-tuned patience rule and to a linear gate trained on the same labels, so its distinctive value is an exact, readable, non-linear decision function rather than higher accuracy. In closed-loop control of 12 held-out live runs, including real learning-rate changes applied to running processes, the full system used 37.1% fewer epochs with no significant change in best validation loss ($p = 0.95$). We also report a failure mode: the gate can repeatedly halve the learning rate, which an intervention cooldown should address.

The study uses one small CNN and one image dataset, and extending it to larger vision backbones and datasets is the main direction for future work. The system is open source and all reported numbers are reproducible from the released scripts.

Keywords: Deep learning; Image classification; Kolmogorov-Arnold networks; Interpretable AI; Early stopping; Training control